apply to discharges led through the shell pla ting from spaces above the bulkhead deck of passenger ships and the freeboard deck of cargo ships.

8.3 Machinery space, main and auxiliary sea inlets and discharges in connection with the operation of machinery shall be fitted with readily accessible valves between the pipes and the shell plating or betw een the pipes and fabricated boxes attached to the shell plating. In manned machinery spaces the valves may be controlled locally and shall be provided with indicators showing whether they are open or closed.

8.4 Moving parts penetrating the shell plating below the deepest subdivision draught shall be fitted with a watert ight sealing arrangement acceptable to the Administration. The inboard gland shall be located within a watertight space of such volume that, if flooded, the bulkhead deck will not be submer ged. The Administration may require that

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SOURCE: MSC.194(80).pdf
The inboard gland shall be located within a watertight space of such volume that, if flooded, the bulkhead deck will not be submer ged. The Administration may require that if such compartment is flooded, essential or emergency power and lighting, internal communication, signals or other emergency devices must remain available in other parts of the ship.

8.5 All shell fittings and valves required by th is regulation shall be of steel, bronze or other approved ductile material. Valves of or dinary cast iron or similar material are not acceptable. All pipes to which this regulation re fers shall be of steel or other equivalent material to the satisfaction of the Administration.

9 Gangway, cargo and fuelling ports fitted below the bulkhead deck of passenger ships and the freeboard deck of cargo ships shal l be watertight and in no case be so fitted as to have their lowest point below the deepest subdivision draught.
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-

10.1 The inboard opening of each ash-chute, r ubbish-chute, etc., shall be fitted with an

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SOURCE: MSC.194(80).pdf
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-
39 -

10.1 The inboard opening of each ash-chute, r ubbish-chute, etc., shall be fitted with an efficient cover.

10.2 If the inboard opening is situated below the bulkhead deck of passenger ships and the freeboard deck of cargo ships, the cover shall be watertight and, in addition, an automatic non-return valve shall be fitted in the chute in an easily accessible position

above the deepest subdivision draught.

Regulation 15-1
External openings in cargo ships

1 All external openings leading to compar tments assumed intact in the damage analysis, which are below the final damage waterline, are required to be watertight.

2 External openings required to be waterti ght in accordance with paragraph 1 shall, except for cargo hatch covers, be fitted with indicators on the bridge.

3 Openings in the shell plati ng below the deck limiting the vertical extent of damage shall be fitted with a device that prevents unauthorized opening if they are accessible during the voyage.

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SOURCE: MSC.194(80).pdf
3 Openings in the shell plati ng below the deck limiting the vertical extent of damage shall be fitted with a device that prevents unauthorized opening if they are accessible during the voyage.

4 Other closing appliances which are kept permanently closed at sea to ensure the watertight integrity of external openings sh all be provided with a notice affixed to each appliance to the effect that it is to be kept closed. Manholes fitted with closely bolted covers need not be so marked.

Regulation 16
Construction and initial tests of watertight doors, sidescuttles, etc.

1 In all ships:

.1 the design, materials and construction of all watertight doors, sidescuttles, gangway and cargo ports, valves, pipes, ash-chut es and rubbish-chutes referred to in these regulations shall be to the satisfaction of the Administration;

.2 such valves, doors and mechanisms sha ll be suitably marked to ensure that they may be properly used to provide maximum safety; and

.3 the frames of vertical watertight doors shall have no groove at the bottom in which dirt might lodge and prevent the door closing properly.

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SOURCE: MSC.194(80).pdf
they may be properly used to provide maximum safety; and

.3 the frames of vertical watertight doors shall have no groove at the bottom in which dirt might lodge and prevent the door closing properly.

2 In passenger ships and cargo ships wate rtight doors shall be tested by water pressure to a head of water they might sustain in a final or intermediate stage of flooding. Where testing of individual doors is not ca rried out because of possible damage to

insulation or outfitting items, testing of i ndividual doors may be replaced by a prototype pressure test of each type and size of door w ith a test pressure corresponding at least to the head required for the intended location. Th e prototype test shall be carried out before the door is fitted. The in stallation method and procedur e for fitting the door on board shall correspond to that of the prototype te st. When fitted on board, each door shall be checked for proper seating between the bulkhead, the frame and the door.

RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-40 -

Regulation 16-1

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SOURCE: MSC.194(80).pdf
checked for proper seating between the bulkhead, the frame and the door.

RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-40 -

Regulation 16-1
Construction and initial tests of watertight decks, trunks, etc.

1 Watertight decks, trunks, tunnels, duct k eels and ventilators shall be of the same strength as watertight bulkheads at corres ponding levels. The means used for making them watertight, and the arrangements adopted for closing openings in them, shall be to the satisfaction of the Administ ration. Watertight ventilator s and trunks shall be carried at least up to the bulkhead deck in passenger ships and up to the freeboard deck in cargo ships.

2 Where a ventilation trunk passing thr ough a structure penetrates the bulkhead deck, the trunk shall be capable of withstandi ng the water pressure that may be present within the trunk, after havi ng taken into account the maximum heel angle allowable during intermediate stages of flooding, in accordance with regulation 7-2.

3 Where all or part of the penetration of the bulkhead deck is on the main ro-ro

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SOURCE: MSC.194(80).pdf
within the trunk, after havi ng taken into account the maximum heel angle allowable during intermediate stages of flooding, in accordance with regulation 7-2.

3 Where all or part of the penetration of the bulkhead deck is on the main ro-ro deck, the trunk shall be capable of withstandi ng impact pressure due to internal water motions (sloshing) of water trapped on the ro-ro deck.

4 After completion, a hose or flooding test shall be applied to watertight decks and a hose test to watertight trunks, tunnels and ventilators.

Regulation 17
Internal watertight integrity of passenger ships above the bulkhead deck

1 The Administration may require that all reasonable and practicable measures shall be taken to limit the entry and spread of water above the bulkhead deck. Such measures may include partial bulkheads or webs. When partial watertight bulkheads and webs are fitted on the bulkhead deck, above or in the i mmediate vicinity of watertight bulkheads, they shall have watertight shell and bulkhead de ck connections so as to restrict the flow of water along the deck when the ship is in a heeled damaged condition. Where the

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SOURCE: MSC.194(80).pdf
they shall have watertight shell and bulkhead de ck connections so as to restrict the flow of water along the deck when the ship is in a heeled damaged condition. Where the partial watertight bulkhead does not line up with the bulkhea d below, the bulkhead deck between shall be made effectively watertig ht. Where openings, pipes, scuppers, electric cables etc. are carried through the partial watertight bulkheads or decks within the immersed part of the bulkhead deck, arrangemen ts shall be made to ensure the watertight integrity of the structure above the bulkhead deck.

2 All openings in the exposed weather deck shall have coamings of ample height and strength and shall be provided with efficient means for expeditiously closing them weathertight. Freeing ports, open rails and sc uppers shall be fitted as necessary for rapidly clearing the weather deck of water under all weather conditions.

3 The open end of air pipes terminating within a superstructure shall be at least 1 m